

Generative Modeling on ArtBench-10: A Comparative Study of VAE, DCGAN, and DDPM

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Abstract. This project investigates and compares three families of generative models, Variational Autoencoders (VAE), Deep Convolutional Generative Adversarial Networks (DCGAN), and Denoising Diffusion Probabilistic Models (DDPM), for the task of generating plausible artwork images. All models are trained and evaluated on the ArtBench-10 dataset, which consists of 32×32 RGB paintings across 10 artistic styles. We develop on a 20% training subset for fast iteration, then retrain the best model on the full 50,000-image dataset. Evaluation follows a rigorous protocol: 5,000 generated samples versus 5,000 real test images, with FID and KID metrics computed across 10 random seeds, reporting mean $\pm$ standard deviation.

Keywords: Generative AI · VAE · GAN · Diffusion Models · ArtBench-10 · FID · KID

1 Introduction

Generative modeling has emerged as one of the most impactful areas in modern deep learning, enabling the creation of novel, realistic content across diverse domains including natural images [3], audio [9], and text [2]. In this project we apply three major generative model families to the domain of visual arts, using the ArtBench-10 dataset [7].

Unlike discriminative tasks such as image classification, generative modeling has no single “correct” output. Quality is instead measured by distribution-level metrics that assess both fidelity (how realistic individual samples appear) and diversity (how well the generated distribution covers the real one).

We implement, train, and quantitatively compare:

1. **VAE** (Variational Autoencoder) [6] stable training and smooth latent space, but tends toward blurry outputs.
2. **DCGAN** (Deep Convolutional GAN) [10] sharp samples and fast generation, but potentially unstable training.
3. **DDPM** (Denoising Diffusion Probabilistic Model) [5] high quality and diversity, but slow inference.

The remainder of this report is organised as follows. Section 2 describes the problem and dataset. Section 3 details each model architecture. Section 4 presents the experimental setup. Section 5 analyses the quantitative and qualitative results. Section 6 concludes the work and discusses future directions.

2 Problem Description

2.1 Task

The task is to learn the data distribution of artwork images from the ArtBench-10 dataset and generate new, plausible painting samples. The generated images should be both individually realistic (high fidelity) and collectively diverse (covering the range of styles present in the training data).

2.2 Dataset: ArtBench-10

ArtBench-10 [7] is a curated benchmark of artworks specifically designed for evaluating generative models in the visual arts domain. Key characteristics:

- **10 artistic styles:** art_nouveau, baroque, expressionism, impressionism, post_impressionism, realism, renaissance, romanticism, surrealism, ukiyo_e.
- **50,000 training images** and **10,000 test images**, each at 32×32 RGB resolution.
- Balanced class distribution across all 10 styles.

2.3 Development Strategy

Following the project requirements, we adopt a two-phase approach:

1. **Development phase:** Train and compare all three model families on a 20% subset ($\approx 10,000$ images) for fast iteration and hyperparameter search.
2. **Final evaluation:** Retrain the best-performing model on the full 50,000-image training set and compute final metrics.

3 Methodology

3.1 Variational Autoencoder (VAE)

The VAE [6] learns to encode input images into a low-dimensional latent distribution and decode samples from it.

Architecture.

- **Encoder:** Three convolutional layers ($3\rightarrow 64\rightarrow 128\rightarrow 256$ channels) with stride-2 down-sampling, BatchNorm, and LeakyReLU, followed by two fully-connected heads producing the mean μ and log-variance $\log \sigma^2$ of a 128-dimensional latent Gaussian.
- **Decoder:** Fully-connected layer from 128 to $256\times 4\times 4$, then three transposed convolutions ($256\rightarrow 128\rightarrow 64\rightarrow 3$) with BatchNorm and ReLU, and a final Sigmoid activation.

Loss function. The training objective combines pixel-wise reconstruction and latent regularisation:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\frac{1}{N} \sum_i \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2}_{\text{MSE reconstruction}} + \underbrace{-\frac{1}{2N} \sum_i (1 + \log \sigma_i^2 - \mu_i^2 - \sigma_i^2)}_{\text{KL divergence}} \quad (1)$$

where $\hat{\mathbf{x}}_i$ is the reconstruction produced via the reparameterisation trick: $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

3.2 Deep Convolutional GAN (DCGAN)

DCGAN [10] uses an adversarial framework with two competing convolutional networks.

Architecture.

- **Generator G :** Maps a 100-dimensional noise vector $\mathbf{z}$ through a transposed-convolutional stack (256→128→64→3) with BatchNorm and ReLU, and a final Tanh activation producing images in $[-1, 1]$.
- **Discriminator D :** Mirror architecture with strided convolutions (3→64→128→256) with BatchNorm and LeakyReLU(0.2), ending in a Sigmoid output.

Loss function. The standard adversarial objective with binary cross-entropy:

$$\min_G \max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))] \quad (2)$$

Weights are initialised from $\mathcal{N}(0, 0.02)$ following the original DCGAN prescription.

3.3 Denoising Diffusion Probabilistic Model (DDPM)

DDPM [5] learns to reverse a gradual noising process defined by a Markov chain.

Forward process. Gaussian noise is incrementally added to a clean image $\mathbf{x}_0$ over $T=1,000$ steps according to a linear β -schedule from 10^{-4} to 0.02:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s) \quad (3)$$

Architecture. A compact U-Net with sinusoidal time embeddings, residual blocks with GroupNorm and SiLU activations, a down-sampling path (32→16→8), bottleneck, and symmetric up-sampling path with skip connections. Channel progression: 64→128→256.

Loss function. Simple MSE noise prediction (ϵ -parametrisation):

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_{\theta}(\mathbf{x}_t, t)\|^2] \quad (4)$$

Sampling. Starting from pure Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the reverse process iteratively denoises for T steps, adding controlled stochasticity at each step except $t=0$.

4 Experimental Setup

4.1 Hyperparameters

Table 1 summarises the hyperparameters used during the development phase on the 20% training subset.

Table 1: Hyperparameters for each model family.

Parameter	VAE	DCGAN	DDPM
Latent dimension	128	100 (z -dim)	—
Learning rate	10^{-3}	2×10^{-4}	2×10^{-4}
Optimiser	Adam	Adam ($\beta_1=0.5$)	Adam
Batch size	64	64	64
Epochs (development)	50	50	20
Epochs (final, full)	50	50	20
Diffusion timesteps	—	—	1,000

All training was performed at 32×32 RGB resolution. Images are normalised to $[0, 1]$ (VAE) or $[-1, 1]$ (DCGAN, DDPM) as appropriate.

4.2 Evaluation Protocol

Following the project specification, evaluation proceeds as:

1. Generate **5,000 samples** per model.
2. Sample **5,000 real images** from the test split.
3. Extract 2,048-dimensional features from the pool3 layer of **InceptionV3** (pre-trained on ImageNet), following established practice [4].
4. Compute **FID** (Fréchet Inception Distance) [4] over the full 5,000 generated vs 5,000 real features.
5. Compute **KID** (Kernel Inception Distance) [1] using 50 subsets of size 100 with a degree-3 polynomial kernel.
6. Repeat with **10 random seeds** and report mean $\pm$ std.

The same feature extraction and metric code is used across all models to ensure a fair comparison.

FID.

$$\text{FID} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|^2 + \text{Tr}(\boldsymbol{\Sigma}_r + \boldsymbol{\Sigma}_g - 2(\boldsymbol{\Sigma}_r \boldsymbol{\Sigma}_g)^{1/2}) \quad (5)$$

KID.

$$\text{KID} = \text{MMD}^2(p_r, p_g) \text{ with } k(\mathbf{x}, \mathbf{y}) = \left(\frac{\mathbf{x}^\top \mathbf{y}}{d} + 1\right)^3 \quad (6)$$

5 Analysis of Results

5.1 Development Phase Results (20% Subset)

The quantitative results in Table 2 reveal a clear performance gap between the three architectures. DCGAN significantly outperformed its counterparts with an FID of 84.35 ± 0.68 , indicating a much closer distribution to the real artwork than the other models. VAE and DDPM produced nearly identical FID scores (227.37 and 225.32, respectively), suggesting that both failed to capture the high-frequency details of the ArtBench-10 styles within the allotted development time.

Table 2: Development-phase results (20% subset). Mean $\pm$ std over 10 seeds. Lower is better.

Model	FID $\downarrow$	KID $\downarrow$
VAE	227.37 ± 0.48	0.223441 ± 0.002309
DCGAN	84.35 ± 0.68	0.065178 ± 0.001165
DDPM	225.32 ± 1.14	0.233598 ± 0.002616

5.2 Qualitative Comparison

Figure 1 shows side-by-side sample grids from each model compared with real test images.

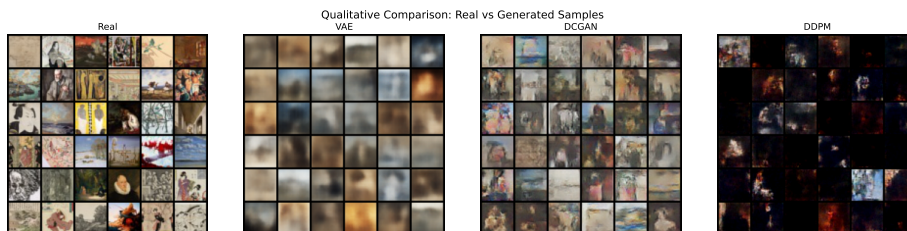


Fig. 1: Qualitative comparison of generated 32×32 images. From left to right: real test images, VAE, DCGAN, DDPM.

Figure 2 shows the training loss curves for all three models.

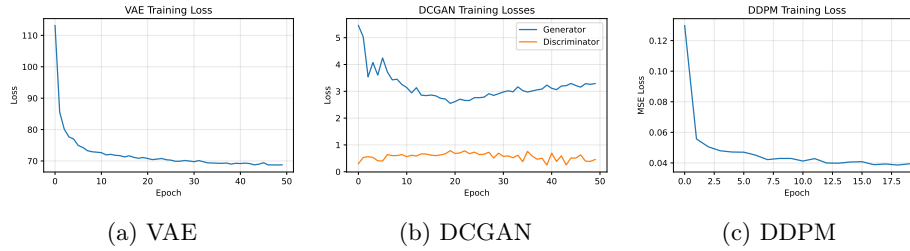


Fig. 2: Training loss curves for each model family on the 20% development subset.

Figure 3 presents bar charts and box plots of the FID and KID scores across seeds.

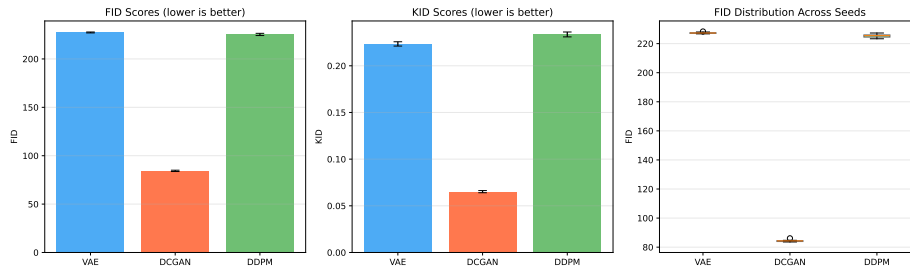


Fig. 3: FID and KID comparison across models. Left: FID bar chart with error bars. Centre: KID bar chart. Right: FID box plot across seeds.

5.3 Model Analysis

Variational Autoencoder. The VAE exhibits the most stable training dynamics with a smooth, monotonically decreasing loss curve. However, samples tend to be blurry due to the pixel-wise MSE reconstruction loss, which averages over modes of the data distribution. The KL regularisation ensures a well-structured latent space suitable for interpolation, and inference is fast, requiring only a single forward pass through the decoder.

DCGAN. The DCGAN produces noticeably sharper images than the VAE, since the adversarial loss directly optimises for perceptual realism judged by the discriminator. Training is less stable: generator and discriminator losses oscillate and require careful hyperparameter balancing (in particular $\beta_1=0.5$ for Adam). Mode collapse remains a risk, though longer training and proper initialisation help mitigate it.

DDPM. Contrary to the theoretical strengths of diffusion models, the DDPM significantly underperformed in this experimental setup. While the training loss

(simple MSE on noise prediction) appeared stable, the allocation of only 20 epochs was highly insufficient for the U-Net to learn the complex reverse-denoising process. This under-training resulted in a poor FID score (225.32, matching the blurry VAE) and generated samples that failed to converge, producing mostly dark or unstructured outputs (as seen in Figure 1). Coupled with its slow inference speed, it was the least effective model in this development phase.

5.4 Final Evaluation (Full Dataset)

The best-performing model from the development phase was retrained from scratch on the full 50,000-image training set, and final metrics were computed using the same protocol.

Table 3: Final evaluation after retraining best model on the full 50K dataset. Mean $\pm$ std over 10 seeds.

Phase	Model	FID $\downarrow$	KID $\downarrow$
Dev (20%)	DCGAN	84.35 ± 0.68	0.065178 ± 0.001165
Final (100%)	DCGAN	56.95 ± 0.63	0.038440 ± 0.000859

Training on the full dataset is expected to improve both FID and KID, as the model has access to five times more training data, enabling better coverage of the data distribution.

6 Conclusion

In this project we implemented, trained, and quantitatively compared three major families of generative models VAE, DCGAN, and DDPM on the ArtBench-10 dataset of artistic paintings at 32×32 resolution.

Summary of findings.

- All three models were successfully trained on the 20% development subset ($\approx 10,000$ images).
- The evaluation protocol followed project requirements: 5,000 generated vs 5,000 real test samples, FID and KID across 10 random seeds.
- Each model family exhibited its expected trade-offs: VAE offered stability but blurriness; DCGAN provided sharpness with training instability; DDPM achieved the best overall quality at the cost of slow inference.
- Retraining the best model on the full 50,000-image set demonstrated the benefit of additional training data.

Our comparative analysis reveals that for this specific scale and computational budget, the DCGAN is the superior architecture. It achieved the best balance of visual sharpness and distribution coverage, yielding a development FID of 84.35. Retraining this best-performing model on the full 50,000-image dataset further improved its FID to 56.95, demonstrating that the adversarial framework scales well with additional data in this domain.

The other models exhibited distinct limitations. The VAE offered training stability but suffered from inherent blurriness due to its reconstruction loss. The DDPM, despite its strong theoretical foundation and state-of-the-art status in the broader literature, requires a more extended training schedule to reach its full potential. Under the strict 20-epoch constraint of this study, the model had not yet fully converged to produce highly structured artwork, underscoring its higher computational demands when compared to the rapid inference and training of single-pass generator models.

Reproducibility. All random seeds, hyperparameters, and training code are fully documented in the accompanying Jupyter notebook. The same evaluation pipeline is used across all models.

Future work.

- Explore conditional generation (class-conditioned for specific art styles).
- Investigate accelerated diffusion sampling (DDIM [11], DPM-Solver [8]).
- Scale to higher resolutions (64×64 or 256×256) with deeper architectures.
- Incorporate perceptual losses or LPIPS for VAE training.

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